Manuel Del Piano

Postdoctoral Researcher

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Born 24 January 1997, Naples · Italian citizen

I use black holes as laboratories for fundamental physics. I build theory-agnostic descriptions of quantum-corrected black hole geometries and carry them through to observables such as shadows and quasi-normal spectra, and I derive analytic descriptions of clouds of ultralight bosons around black holes, the backgrounds needed to search for axion-like particles with gravitational waves. Ten papers since 2022, all published in Physical Review D and the European Physical Journal C.

Current position

Postdoctoral Researcher, QTC, University of Southern Denmark, Odense — Nov 2025 – present
Group of Prof. F. Sannino: quantum gravity phenomenology, effective field theory of black holes, non-perturbative effects of strong interactions.

Collaborations

Core Member, LISA Consortium, Fundamental Physics and Waveform Working Groups — 2026 – present
Contributing to the projects *EMRI models with axion clouds* and *The Natural Log Project: technically natural quantum black hole signatures with LISA*.

Member, COST Action CA23130, BridgeQG — Nov 2024 – present

Member, Quantum Field Theory Center (QTC), SDU — Nov 2023 – present

Member, INFN Sezione di Napoli — Nov 2021 – Oct 2025

Education

PhD in Physics, Scuola Superiore Meridionale and INFN Sezione di Napoli — Nov 2021 – Dec 2025
Excellent with honours, 17 December 2025. Thesis: *The Effective Metric Descriptions: a parametrization for black hole spacetimes beyond general relativity*.

MSc in Physics, Università degli Studi di Napoli "Federico II" — Sep 2019 – Sep 2021
110/110 *cum laude*. Thesis: *Charting Swampland with Black Hole Physics*.

BSc in Physics, Università degli Studi di Napoli "Federico II" — Sep 2016 – Oct 2019
110/110 *cum laude*. Thesis: *Testing the Einstein Equivalence Principle with Space Experiments*.

Diploma di Maturità Scientifica, Scuola Militare "Nunziatella", Naples — Sep 2013 – Jul 2016

Publications

Peer-reviewed

[1] M. de Cesare, M. Del Piano, C. A. R. Herdeiro, *Analytic backreaction of a scalar wig on a Schwarzschild black hole*, Phys. Rev. D **114** (2026) 084059, arXiv:2607.25932

[2] G. Cacciapaglia, M. Del Piano, F. Sannino, V. Vellucci, *When black holes admit pair-of-pants topologies*, Phys. Rev. D **114** (2026) 084072, arXiv:2606.24642; selected for APS press outreach

[3] M. Del Piano, C. De Simone, M. Damia Paciarini, V. De Falco, M. Myszkowski, F. Sannino, V. Vellucci, *Toward a unified view of agnostic parametrizations for deformed black holes*, Phys. Rev. D **113** (2026) 064010, arXiv:2510.14707

[4] A. Alonso-Serrano, M. de Cesare, M. Del Piano, *Black hole solutions in quantum phenomenological gravitational dynamics*, Phys. Rev. D **112** (2025) 084031, arXiv:2504.01539

[5] M. Damia Paciarini, M. Del Piano, S. Hohenegger, F. Sannino, *Effective metric description of charged black holes*, Eur. Phys. J. C **85** (2025) 848, arXiv:2504.20810

[6] M. Del Piano, S. Hohenegger, F. Sannino, *Black hole shadow and other observables away from the horizon: extending the effective metric descriptions*, Phys. Rev. D **111** (2025) 064070, arXiv:2412.13673

[7] M. Del Piano, S. Hohenegger, F. Sannino, *Effective metric descriptions of quantum black holes*, Eur. Phys. J. C **84** (2024) 1273, arXiv:2403.12679

[8] B. Bajc, M. Del Piano, F. Sannino, *UV finite GUT with SUSY breaking*, Phys. Rev. D **109** (2024) 055043, arXiv:2308.13311

[9] M. Del Piano, S. Hohenegger, F. Sannino, *Quantum black hole physics from the event horizon*, Phys. Rev. D **109** (2024) 024045, arXiv:2307.13489

[10] E. Binetti, M. Del Piano, S. Hohenegger, F. Pezzella, F. Sannino, *The effective theory of quantum black holes*, Phys. Rev. D **106** (2022) 046006, arXiv:2203.13515

In preparation

Semiclassical scaling dimensions of gauge-invariant operators in pure Yang–Mills theory.

Invited seminars

QTC Seminar, Quantum Field Theory Center, University of Southern Denmark, Odense, *Analytic backreaction of a scalar wig on a Schwarzschild black hole* Oct 2026

CEICO Cosmology Seminar, Czech Academy of Sciences, Prague, *Effective Metric Description for Quantum Black Holes* Feb 2026

Quantum and Classical Gravity Group, Humboldt University, Berlin (host: A. Alonso-Serrano), on Effective Metric Descriptions May 2025

Strong Group Seminar, Niels Bohr Institute, Copenhagen (host: V. Cardoso), *An Effective Metric Description for Quantum Black Holes* Mar 2025

ICC Seminar, Institute of Cosmos Sciences, University of Barcelona (host: R. Emparan), *Quantum black hole physics from the event horizon* Oct 2024

Conference contributions

From Black Holes to the Cosmos: Matt Visser's journey through space and time, SISSA, Trieste, *poster* on the scalar-wig backreaction Aug 2026

PAFT26, Quantum Frontiers in Gravity, Vietri sul Mare, Italy Mar–Apr 2026

XXVI SIGRAV Conference on General Relativity and Gravitation, Milan, *talk* Sep 2025

Astrophysics and Space Science in Marche II, Camerino, Italy, *talk* Sep 2025

Quantum Phases of Strong Dynamics, Odense, Denmark Aug 2025

BridgeQG First Annual Meeting, COST Action CA23130, Paris Jul 2025

Mathematical Physics of Gravity and Symmetry, ENS de Lyon, *talk* Jun 2025

Theories of Fundamental Interactions (TFI 2024), Naples, *poster* Sep 2024

PAFT 2024, Vietri sul Mare, Italy, *talk* Mar 2024

New Horizons on Primordial Black Holes (NEHOP23), Naples, *talk* Jun 2023

Teaching

Teaching Assistant, Quantum Mechanics II, SDU, theory and exercises 2026

Private tutor in mathematics and physics, secondary-school and undergraduate level 2020 – present

Academic service and organisation

Managing Editor (part-time), Annals of Physics from Nov 2026

Pre-editorial checks of submissions for novelty and ethical compliance.

Referee (since Oct 2024) for Classical and Quantum Gravity, JCAP, European Physical Journal C, Annals of Physics, Physics of the Dark Universe, and the International Journal of Geometric Methods in Modern Physics.

Organiser, BIFROST: Black hole Insights, navigating the FROntiers of SpaceTime, Odense, Oct 2025.

Awards and recognition

APS press outreach: paper [2] selected by the American Physical Society for its outreach to the press, 2026.

Scholarship from the Nunziatella Alumni Association in support of undergraduate studies.

Languages and computational skills

Languages. Italian (native), English (C1), Danish (A2), French (elementary).

Computing. Mathematica, Python, MATLAB, C, C++, Linux, LabVIEW, SRIM, LaTeX.

Methods. Spectral and continued-fraction (Leaver) methods for quasi-normal and quasi-bound spectra; matched asymptotic expansions; symbolic tensor computation.

Design and media. Adobe Illustrator, Photoshop, Premiere, GIMP, Blender, Autodesk Fusion 360.